INNOVATOR TEST 5
MATH

MODULE 1

1. A tree-cutting service has a price plan that consists of a one-time fee of $\$600$ to come out to a site, plus a fee of $\$200$ for each tree the service cuts down at the site. A graph of a line with a positive slope and a positive y–intercept is used to represent this information. What is the best interpretation of the slope of this graph?

PREV. QUESTION | NEXT QUESTION

- A) The one-time fee
- B) An hourly rate for the time the service is onsite
- C) The fee for all of the trees cut down on a site
- D) The fee for each tree cut down at the site

2. If $\frac{c}{11} = 8$, what is the value of $\frac{11}{c}$?

3. $3, 3, 5, 5, 5, 5, 7, 10, 10$

Vira recorded the distances of her recent jogs, in miles, in the data list. Which frequency table accurately represents this data?

PREV. QUESTION | NEXT QUESTION

A

Distance	Frequency
3	2
5	6
7	7
10	9

B

Distance	Frequency
3	2
5	4
7	1
10	2

C

Distance	Frequency
1	7
2	3
2	10
4	5

D

Distance	Frequency
6	2
7	1
20	2
20	4

4. Which of the following expressions is equivalent to $(a^6b^7c^{-2})(a^2bc^5)$?

PREV. QUESTION | NEXT QUESTION

A $a^{12}b^7c^{-10}$

B $a^8b^{12}c^3$

C $a^8b^8c^3$

D $a^{36}b^7c^{-32}$

5. Which of the following systems of equations has no solutions?

PREV. QUESTION | NEXT QUESTION

A $y = 3x + 5$
$y = 4x + 5$

B $y = 7$
$y = 2x + 11$

C $y = 7$
$x = 12$

D $y = 8x + 1$
$y = 8x + 13$

6. A rectangular solid has a square base with edges of length 23 inches and a height of 17 inches. What is the volume, in cubic inches, of the rectangular solid?

PREV. QUESTION | NEXT QUESTION

A 126

B $4{,}913$

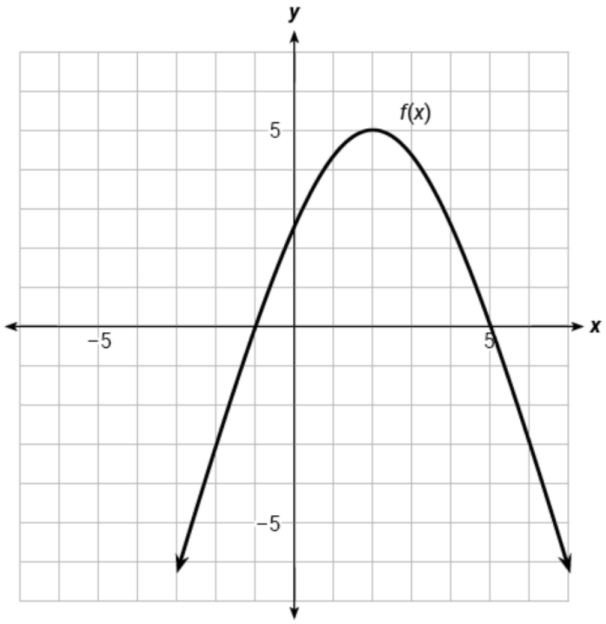
C $8{,}993$

D $12{,}167$

7. $$15R + 70B = 465$$

After an evening at an arcade, Vivian works out an equation to represent how many plastic rings, R, and how many bouncy balls, B, she can get with the 465 tickets she won playing arcade games. The equation represents this situation. The cost of a bouncy ball is how many more tickets than that of a plastic ring?

8.



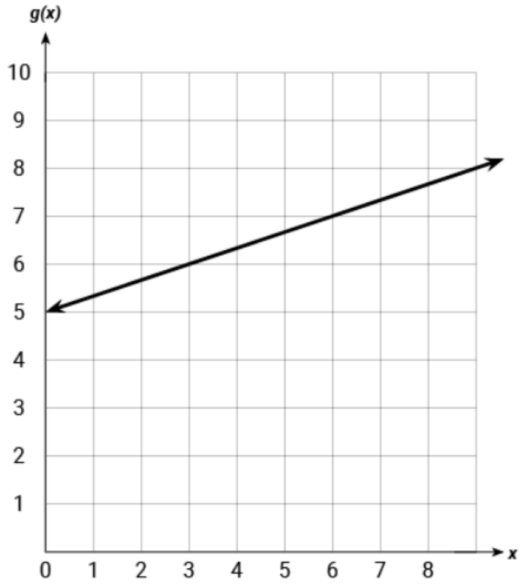
The graph of the function $f(x)$ is shown. What value(s) of x satisfy $f(x) = 0$.

PREV. QUESTION | NEXT QUESTION

- A 2.5
- B 3.5
- C -1 and 5
- D There are no values of x where $f(x) = 0$.

COME BACK LATER

9.



The graph of $g(x)$ is shown. What is the value of $g(6)$?

PREV. QUESTION | NEXT QUESTION

- A 3
- B 6
- C 7
- D 8

COME BACK LATER

10.

	Red Team	White Team	Blue Team	Total
7th Grader	32	36	35	103
8th Grader	40	34	43	117
Total	72	70	78	220

At North Shore Junior High School, students are either in 7th grade or 8th grade. All students are on either the red team, white team, or blue team for assemblies, mentoring, and intramural sports. The table summarizes the distribution of students. If one student is picked at random from all of the students at the school to ride in the lead car of the school's annual Oyster Day parade, what is the probability that the student will be on the white team?

11. $\frac{x+8}{3} = \frac{16}{x}$

What is the positive solution to the equation above?

12. As part of a mark-recapture study, a random sample of cricket frogs was selected from a previously surveyed population to estimate what percentage of the population had already been tagged. Based on the sample, it was estimated that 68% of the entire population of cricket frogs in the study region had been tagged, with an associated margin of error of 3%. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the percentage of the population that was tagged?

PREV. QUESTION NEXT QUESTION

× A It is plausible that the percentage is less than 65%.

× B The percentage is exactly 68%.

× C It is plausible that the percentage is greater than 71%.

× D It is plausible that the percentage is between 65% and 71%.

13. The possible values of a number w are at least 38 less than twice another number v. Which inequality shows all possible values of w?

PREV. QUESTION | NEXT QUESTION

× A $w \geq 38 - 2v$

× B $w \leq 38 - 2v$

× C $w \geq 2v - 38$

× D $w \leq 2v - 38$

14. $$F = \frac{Gm_1m_2}{r^2}$$

The equation above defines the force, in newtons, of gravity between two objects in terms of the masses of the two objects m_1 and m_2 and the distance between the two objects, r. G is the universal gravitational constant.

Which equation correctly expresses r^2 in terms of G, m_1, m_2, and F?

PREV. QUESTION | NEXT QUESTION

× A $r^2 = \frac{m_1m_2}{GF}$

× B $r^2 = \frac{Gm_1m_2}{F}$

× C $r^2 = \frac{F}{Gm_1m_2}$

× D $r^2 = FGm_1m_2$

15. The graph of a line in the xy-plane contains the points $(-3, -14)$ and $(0, 4)$. Which of the following is the equation of the line?

PREV. QUESTION | NEXT QUESTION

× A $y = 3x + 4$

× B $y = 3x + 18$

× C $y = 6x + 4$

× D $y = 18x + 3$

16. A local aquarium is going to increase the volume of its cylindrical tank by 50%. If the radius is increased by 18%, the height of the tank would have to be increased by what percent (rounded to the nearest tenth of a percent)?

PREV. QUESTION | NEXT QUESTION

- A) 2.0%
- B) 7.7%
- C) 10.3%
- D) 32%

17. Right triangle MNO has side lengths given by the following: $MN = 24$, $NO = 45$, and $OM = 51$. Triangle PQR is similar to triangle MNO. What is the value of tan R?

PREV. QUESTION | NEXT QUESTION

- A) $\frac{8}{17}$
- B) $\frac{8}{15}$
- C) $\frac{15}{17}$
- D) $\frac{15}{8}$

18. Mario is cutting a pipe that is 66 inches long into three pieces. The second piece is 50 percent longer than the first, and the third piece is twice as long as the second. How long, in inches, is the longest piece?

PREV. QUESTION | NEXT QUESTION

- A) 12
- B) 18
- C) 24
- D) 36

19. The graph of $x^2 + y^2 + 3x + 5y = \frac{15}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

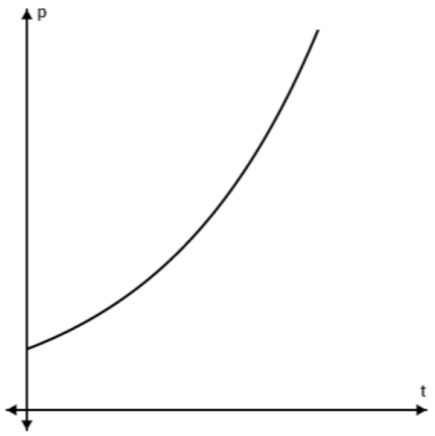
20. $$y = 2x^2 - 12x + 14$$
$$y = -4$$

The given system of equations has exactly one solution at $(a, -4)$. What is the value of a?

PREV. QUESTION | NEXT QUESTION

- A -4
- B -3
- C 3
- D 6

21.



A researcher models the population of lizards on an island using the equation $p(t) = 80(1 + a)^t$, where t is the number of years after the initial observation and $0 < a < 1$. The partial graph of $p(t)$ is shown. What is the best interpretation of a in this context?

PREV. QUESTION | NEXT QUESTION

A. It takes a days in the year for the population to reach 80.

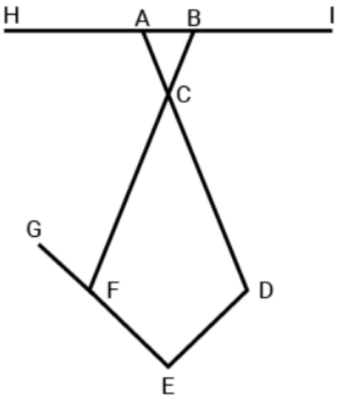
B. The population of lizards will increase by 80 every a years.

C. The population of lizards will increase by $100a$ percent each year.

D. The population of lizards will decrease by $100a$ percent each year.

COME BACK LATER

22.



Note: Figure not drawn to scale.

In the figure shown, triangle ABC is an isosceles triangle, the length of CF is equal to the length of CD, and the length of FE is equal to the length of DE. If the measure of angle HAC is 100° and the measure of angle GFC is 40°, what is the measure of angle E, in degrees?

PREV. QUESTION | NEXT QUESTION

A. 20

B. 60

C. 80

D. 140

COME BACK LATER

MODULE 2

1. $$h(t) = 80{,}000(1 - \frac{12}{100})^t$$

Leonardo just bought a new boat, and the value of the boat depreciates at a rate of 12% per year. The function $h(t)$ gives the value of his boat, in dollars, after t years have passed. Which of the following expressions could be used to determine the number of years it would take for the value of his boat to decrease by 25%?

PREV. QUESTION | NEXT QUESTION

- A $20{,}000 = 80{,}000(1 - \frac{12}{100})^t$
- B $20{,}000 = 80{,}000(1 - \frac{25}{100})^t$
- C $60{,}000 = 80{,}000(1 - \frac{12}{100})^t$
- D $60{,}000 = 80{,}000(1 - \frac{12}{100})^{4t}$

2. $$y = -\frac{2}{3}x + 4$$
$$y = -\frac{2}{3}x - 11$$

At how many points do the graphs of the given equations intersect in the xy-plane?

PREV. QUESTION | NEXT QUESTION

- A Zero
- B Exactly one
- C Exactly two
- D Infinitely many

3. A partially filled gas tank contains 2 gallons of gasoline. If a fuel pump can pump at 0.5 gallons per second and the maximum capacity of the tank is 14 gallons, after how much time, in seconds, will the tank be full?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

4. From 1994 to 1995, the amount of money, in dollars, a company earns grew by 6%. Each year for the next ten years, the amount of money the company earns is equal to the amount earned between 1994 and 1995. What type of function best models the relationship between the money the company earns, in dollars, and time, in years?

PREV. QUESTION | NEXT QUESTION

- A Decreasing linear
- B Decreasing exponential
- C Increasing linear
- D Increasing exponential

5. If $\frac{p}{q} = \frac{4}{3}$, what is the value of $\frac{12q}{p}$?

PREV. QUESTION | NEXT QUESTION

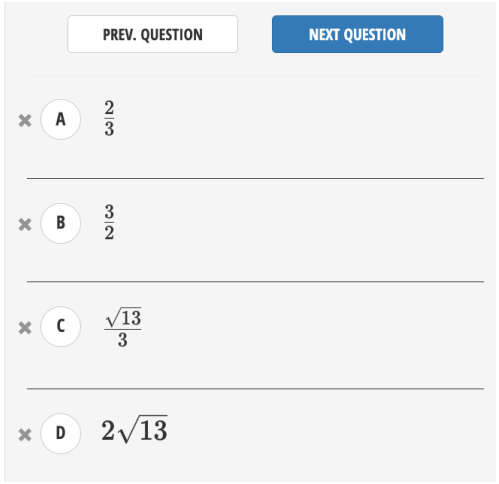
- A 6
- B 8
- C 9
- D 16

6. The exponential function f is defined by $f(x) = 11 \cdot b^x$, where b is a positive constant. If $f(4) = 2{,}816$, what is the value of $f(5)$?

PREV. QUESTION | NEXT QUESTION

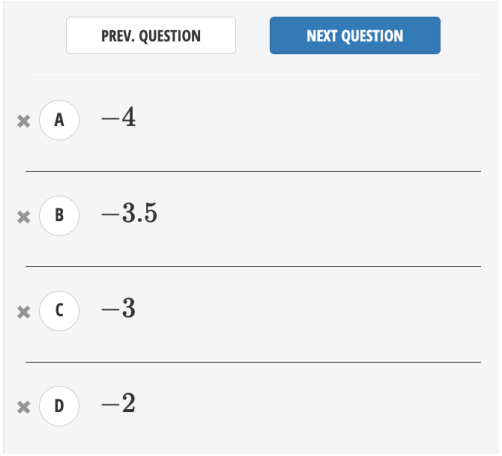
- A 4
- B $11{,}264$
- C $161{,}051$
- D $164{,}916{,}224$

7. What is the sum of all values of b that satisfy the equation $3b^2 - 2b - 4 = 0$?

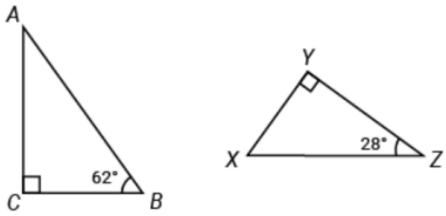


8. $$f(x) = mx + 4$$
$$g(x) = \frac{1}{2}x - 5$$

If $f(2) = g(4)$, what is the value of m ?



9.



Note: Figure not drawn to scale.

Triangles ABC and XYZ are shown. If $\frac{AB}{BC} = k$, where k is a constant, which of the following ratios is equal to k?

PREV. QUESTION | NEXT QUESTION

- A $\frac{XZ}{XY}$
- B $\frac{XY}{XZ}$
- C $\frac{YZ}{XZ}$
- D $\frac{XZ}{YZ}$

10. $(ax + 4)(3x + b) = 18x^2 + cx + 12$

If the equation is defined for all values of x, what is the value of c?

PREV. QUESTION | NEXT QUESTION

- A 3
- B 6
- C 15
- D 30

11. The equation $x^2 + y^2 = 4x - 6y + 23$ defines a circle in the xy-plane. What is the length of the radius of the circle?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

12. If $x^3 - 7x^2 + 3x - 21 = 0$, what is a real number solution for x?

PREV. QUESTION NEXT QUESTION

13. Square $ABCD$ has an area of 100 square units. If E, F, G and H are all midpoints of the sides of $ABCD$, respectively, what is the perimeter, in units, of square $EFGH$?

PREV. QUESTION NEXT QUESTION

- A $5\sqrt{2}$
- B 20
- C $20\sqrt{2}$
- D $40\sqrt{2}$

14. $$h(x) = \frac{1}{(x+3)^2+(x+3)-42}$$

For what value of $x > 0$ is the function $h(x)$ undefined?

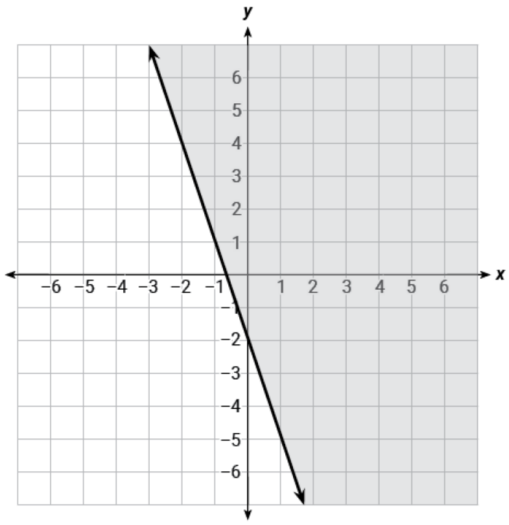
PREV. QUESTION NEXT QUESTION

15. The variables x and y are related such that each time x increases by 4, y decreases by 3. If the function goes through the point $(8, -5)$, which of the equations below represents the function?

PREV. QUESTION | NEXT QUESTION

- A $3x + 4y = -5$
- B $4x - 3y = 1$
- C $4x - 3y = 8$
- D $3x + 4y = 4$

16.



Which of the following inequalities represents the shaded region in the given graph?

PREV. QUESTION | NEXT QUESTION

- A $y \leq -2x - 2$
- B $y \leq -3x - 2$
- C $y \geq -2x - 2$
- D $y \geq -3x - 2$

COME BACK LATER

17.

$$f(x) = mx + b$$

$$g(x) = 2x^2 - 8x + 13$$

In the given system of equations, m and b are constants, $f(x)$ intersects $g(x)$ at the minimum of $g(x)$, and the functions have the same y–intercept. What is the value of m?

PREV. QUESTION | NEXT QUESTION

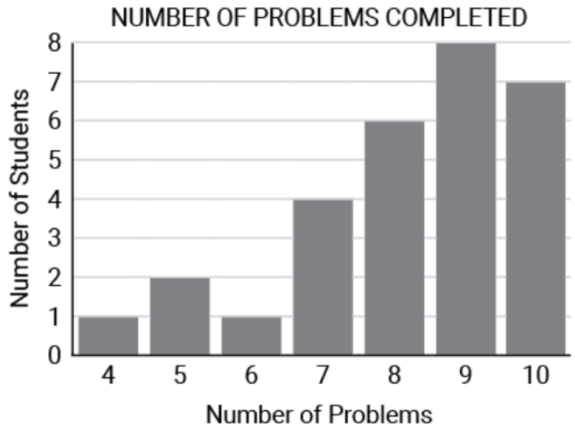
- A -4
- B -2
- C 5
- D 13

18. Line m passes through the points $(a, 0)$ and $(0, c)$, where a and c are real numbers that do not equal 0. If line n is perpendicular to line m and the lines intersect at $(0, c)$, what is the equation of line n?

PREV. QUESTION | NEXT QUESTION

- A $y = -\frac{a}{c}x + c$
- B $y = -\frac{c}{a}x + c$
- C $y = \frac{a}{c}x + c$
- D $y = \frac{c}{a}x + c$

19.



Mr. Kim collected the homework from his students and made a graph to show how many problems his students were able to complete. One of his students turned in their homework late and they completed 10 problems. Which of the following describes what will happen to the data when that data point is added?

PREV. QUESTION | NEXT QUESTION

- A. The mean and median will not change.
- B. The mean will increase slightly and the median will not change.
- C. The mean will not change and the median will increase slightly.
- D. The mean and median will increase slightly.

COME BACK LATER

20. A circle graphed on the coordinate plane is centered at the origin, labeled point O. Point X lies on the circle at $(-k, k\sqrt{3})$. Point Y lies on the circle at $(-2k, 0)$. Which of the following is the measure of angle XOY?

PREV. QUESTION | NEXT QUESTION

- A. $\frac{\pi}{6}$
- B. $\frac{\pi}{3}$
- C. $\frac{2\pi}{3}$
- D. $\frac{5\pi}{6}$

21. The force of gravity on Earth is 9.8 meters per second squared. The gravitational force on Mars is 62% less than that on Earth. What is the approximate force of gravity on Mars, in kilometers per minute squared?

PREV. QUESTION | NEXT QUESTION

- A 0.22
- B 0.36
- C 13.41
- D 21.87

22. A is the first of six consecutive odd integers. If the product of 12 and the 3rd integer is at most 18 less than the sum of the 4th and 6th integers, what is the greatest value of A?

PREV. QUESTION | NEXT QUESTION

- A -6
- B -5
- C 3
- D 11